Elhanan Buchsbaum

Software & Data Systems · Technical Project Delivery · AI-Assisted Development

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Summary

Computational scientist (PhD) who builds software and hardware systems end to end — defining requirements and acceptance criteria up front, implementing and validating the solution, then documenting and training others to run it independently. Nine years writing production research software in Python and Rust, across data pipelines, HPC infrastructure, computer vision, and applied AI/LLM-assisted development tooling (Claude Code, MCP-based tool integrations). Track record of modernizing legacy tooling that was then adopted team-wide, and of translating between technical implementation and non-technical stakeholders. Two first-author Current Biology papers. Native English speaker; German B1–B2.

Experience

Postdoctoral Researcher, Max Planck Institute for Neurobiology of Behavior (caesar) – Bonn, Germany — Jan 2025 – Jan 2026

- Built imaging hardware (camera acquisition, lighting control, custom mechanical mount) and a Python/OpenCV/FFmpeg pipeline with a YOLO model to automate live/dead nematode scoring, replacing slow manual assessment — prototyped end to end within a short three-month contract; the classification model reached notebook-stage validation.

Doctoral Researcher, MPI for Neurobiology of Behavior (caesar) / University of Bonn – Bonn, Germany — 2019 – 2025

- Designed and built a closed-loop measurement platform for freely flying Drosophila end to end; defined datasets, test setups, and acceptance criteria up front, then implemented multi-camera imaging, real-time 3D tracking, optogenetic control, and microcontroller triggering, synchronised at 10 ms closed-loop latency.
- Evaluated Python against the platform's timing requirements, found it insufficient, and rewrote the acquisition/triggering layer in Rust; kept Python for surrounding tooling and analysis.
- Built and validated the multi-camera 3D calibration stack quantitatively (0.5 px reprojection error across 6 cameras at 100 fps).
- Developed reproducible Python pipelines processing hundreds of recordings (~10 GB each, thousands of animals) in HDF5/Parquet, run in parallel locally and as Slurm batch jobs.
- Documented the system and shipped self-service analysis tools so lab members without programming backgrounds could run standardised experiments and analyses unsupervised; trained and supported them. Supervised one MSc student through their thesis project.
- **Result:** first-author paper in Current Biology (2025) on a descending neuron driving visually evoked flight saccades.

MSc Researcher, Technion – Israel Institute of Technology, Faculty of Medicine – Haifa, Israel — 2016 – 2018

- Built MATLAB signal-processing pipelines for in vivo electrophysiology and statistical analysis over large multi-dimensional datasets (~2,100 sorted units, 23 animals).
- Modernised and extended the lab's legacy analysis codebase; documented procedures and trained incoming researchers — the tooling was adopted lab-wide.
- Co-first-author paper in Current Biology (2021) on head-direction coding in the avian hippocampal formation.

Teaching Assistant — Machine Learning in Medical Sciences, Technion – Israel Institute of Technology, Faculty of Medicine – Haifa, Israel 2017 – 2018

- Taught Python, statistics, and machine-learning fundamentals to medical students with no programming background.

Selected Projects

OptoFly — closed-loop tracking and optogenetic stimulation system for freely flying Drosophila – public, GPL-3.0 2020 – 2026

- Integrates multi-camera 3D tracking with triggered recording, optogenetic stimulation, dynamic autofocus, and configurable visual stimuli; Rust acquisition layer with Python bindings. 485 commits; documented via a live docs site.

Skills

AI & automation: LLM-assisted software development (Claude Code), MCP-based tool/skill integration, agentic coding harnesses (opencode, pi), OpenAI-compatible APIs, OpenRouter

Software development: Python (primary), Rust, MATLAB, R, Bash, SQL, Git/GitHub

Data & infrastructure: NumPy/pandas/scikit-learn, statistical modelling, HDF5/Parquet, Linux, Docker, HPC (Slurm), reproducible parallel/batch pipelines

Computer vision: OpenCV, FFmpeg, camera calibration, object detection (YOLO), markerless pose tracking (DeepLabCut, SLEAP)

Working methods: Translating requirements into technical solutions and acceptance criteria; iterative, incrementally validated development; technical documentation for non-technical stakeholders; training and onboarding

Productivity tools: Microsoft Office (Word, PowerPoint)

Education

University of Bonn / MPI for Neurobiology of Behavior (caesar), PhD in Neuroscience – Bonn, Germany 2019 – 2025

- magna cum laude
- Thesis: *Neural Control of Flight Maneuvers in Freely Flying Drosophila melanogaster* (advisor: Dr. Bettina Schnell)
- IMPRS for Brain and Behavior

Technion – Israel Institute of Technology, MSc in Neuroscience – Haifa, Israel 2016 – 2018

- summa cum laude
- Thesis: *Space Coding in the Japanese Quail Hippocampal Formation* (advisor: Prof. Yoram Gutfreund)

University of Haifa, BSc in Biology and Psychology – Haifa, Israel 2013 – 2016

Publications

Activity of a descending neuron associated with visually elicited flight saccades in Drosophila 2025
Elhanan Buchsbaum, Bettina Schnell
10.1016/j.cub.2024.12.001 (Current Biology 35(3), 665–671)

Directional tuning in the hippocampal formation of birds 2021

Co-first author (with K. Krivoruchko). Published under the author's former surname, Ben-Yishay.
Elhanan Ben-Yishay, K. Krivoruchko, S. Ron, N. Ulanovsky, D. Derdikman, Y. Gutfreund
10.1016/j.cub.2021.04.029 (Current Biology 31(12), 2592–2602)

Languages

English: Native
Hebrew: Native
German: B1–B2, actively studying